Aadarsha Gopala Reddy

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SUMMARY

Data-driven researcher who pairs Python/SQL data engineering with simulation benchmarking and LLM-based tooling, seeking a research role applying AI to power systems and energy infrastructure.

SKILLS

Programming & Data Analysis: Python, SQL, Pandas, NumPy, R.

AI & LLM Development: Gemini/OpenAI APIs, agentic coding workflows with Claude Code, LLM-assisted workflow automation.

Data Engineering & Simulation: Apache Airflow, Snowflake, Spark, Kafka, Flink, EnergyPlus simulation benchmarking.

Visualization & Tools: Tableau, Power BI, Git, AWS, PostgreSQL.

EXPERIENCE

Washington University in St. Louis — St. Louis, Missouri, USA

Master's Thesis Research — **August 2025 - May 2026**

- Built an end-to-end pipeline processing 790,000 naturalistic driving trips from 304 participants, engineering behavioral features and benchmarking six model architectures under leave-one-participant-out cross-validation for a faculty-reviewed thesis.

Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale — **January 2026 - May 2026**

- Taught large-scale data pipeline concepts (Snowflake, Airflow, Spark, Kafka, Flink) to a graduate cohort, covering fault-tolerant design, streaming windowing, and query optimization.
- Authored instructional documentation and debugged student pipeline failures weekly through hands-on code review and testing feedback.

Crittero, Inc. — Remote

AI Engineer Intern — **June 2025 - August 2025**

- Generated 100,000 labeled training records via time-series behavioral modeling and trained a TensorFlow/Keras recommendation model achieving 80% accuracy, benchmarked against a random baseline and a collaborative-filtering fallback.
- Built a configurable Python CLI for reproducible model evaluation and benchmarking across hyperparameters, embedding dimensions, and fallback thresholds; established Git-based workflow practices in a multi-developer codebase.

Lab714 — Boca Raton, Florida, USA

Data Analyst Intern — **June 2023 - May 2024**

- Wrote SQL queries and built AWS data ingestion pipelines (S3, Lambda) to process, normalize, and aggregate IoT sensor telemetry, powering trend-analytics and anomaly-detection dashboards for real-time operational monitoring.

PROJECTS

Washington University in St. Louis — St. Louis, Missouri, USA

Datacenter Cooling Optimization using Deep RL — **August 2024 - December 2024**

- Implemented DDQN, PPO, and SAC reinforcement learning agents benchmarked against EnergyPlus building-energy simulations, achieving a 35.8% energy efficiency improvement over baseline controllers.

Red-Blue Visual Auto Defender — **August 2025 - December 2025**

- Built an automated red-blue teaming pipeline using Claude Code for agentic coding workflows, generating visual jailbreak attacks against vision-language models and auto-generating deterministic OCR-based Python defenses.

EDUCATION

Washington University in St. Louis — St. Louis, Missouri, USA

M.S. in Computer Science — **August 2024 - May 2026**

- *Thesis:* Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data

Ohio Wesleyan University — Delaware, Ohio, USA

B.A. in Computer Science and Data Analytics — **August 2020 - May 2023**

- *Minor:* Economics | *Honors:* Golden Bishop Award, Mortar Board, Florence Leas Prize, Dean's List